Hochschule für Technik
und Wirtschaft Berlin
University of Applied Sciences

Named Entity Recognition in German-language Telegram channels

Bachelor Thesis

B.Sc.Computer Science and Business Administration

Faculty 4

by

Elisabeth Steffen

Berlin, August 19, 2022

1st Supervisor: Prof. Dr. Helena Mihaljević
2nd Supervisor: Prof. Dr. Theresa Busjahn

Abstract

This thesis investigates the potentials and limitations of Named Entity Recognition (NER) approaches for the automated analysis of text data from the messenger service Telegram.

The thematic focus is on the mobilizations for protests against the COVID-19 measures in Germany. The rise of conspiracy theories in the context of the COVID-19 pandemic not only reinforced antisemitic and racist discrimination against certain communities, but also generated an increasingly toxic and violent attitude towards political decision-makers and scientific experts.

The past months have once again shown the importance of investigative work by critical journalists and non-governmental organizations to critically monitor the corresponding mobilization strategies on Telegram. A central challenge here is the large amounts of user-generated data from often very dynamic contexts, which are usually analyzed manually. Here, approaches from the field of natural language processing can offer a valuable addition.

Against this background, this thesis aims to evaluate the applicability of Named Entity Recognition for the investigation of the COVID-19 protest movement in German-language Telegram channels, focusing on entities representing people, places, organizations, actions, dates, and times.

This work contributes to the development of German-language Named Entity Recognition in general by applying this approach to a specific text genre and domain, and provides first insights for developing an NER tool for researchers and non-governmental organizations in the future.

Contents

Chapter 1

Introduction

1.1 Background

The outbreak of the COVID-19 pandemic was followed by an increasing spread of misinformation and conspiracy theories. Those narratives evolved not only around the virus itself, but also about the intentions of those responsible for putting into effect measures for its containment.

Those measures included the mandatory use of face masks, rapid testing, and vaccinations, at times functioning as admission requirements in the workplace or publicly accessible spaces such as restaurants and theatres. Furthermore, unprecedented restrictions were imposed on public life through several general or partial lock-downs of social and economic life, as well as temporary bans of gathering in public. These restrictions also heavily affected the possibilities to hold political demonstrations.

From the beginning on, these measures were accompanied by critique and protests from various political spectres. In Germany, the first noticeable protests against COVID-19 measures took place at the so-called 'Hygienedemonstrationen' (hygiene demonstrations) in front of the 'Volksbühne' theatre in Berlin. Over time, the so-called 'Querdenken' (lateral thinking) movement became the most prominent movement evolving around protests

against the COVID-19 containment measures. The movement reached its preliminary climax with several large demonstrations taking place in Berlin, directed against legislative changes regarding the containment of the pandemic passed by the German parliament. In the context of these demonstrations, violent clashes occurred between police and protesters, which often refused to act according the official restrictions for the demonstration such as wearing a face mask. Furthermore, there were several incidents in which journalists were violently attacked by protesters, accusing them to be part of the 'Lügenpresse' (lying press), and two incidents in which the seat of the German parliament, the Reichstag, was targeted by protesters.

On April 21st 2021, the last large authorized demonstration of the 'Querdenken' movement took place in Berlin, comparing a new law for the containment of the pandemic with the 'Ermächtigungsgesetz' put into effect in 1933 which ultimately paved the way for the Nazi dictatorship in Germany. In response to several violations of official restrictions by demonstration participants, police decided to disperse the demonstration. Several protesters subsequently moved to other areas nearby. Clashes between protesters and police continued to occur throughout the rest of the day.

The subsequent demonstrations announced by the 'Querdenken' movement to take place in Berlin were prohibited by authorities, arguing with the experience at prior demonstrations that participants would act against against the official restrictions. The ban was confirmed by Berlin's administrative court after the organizers of the protest had filed a suit against the initial prohibition. In spite of the ban, several hundred protesters gathered in different spots in Berlin e.g. on the 21st, 28th and 29th of August 2021.

The mentioned bans of large demonstrations of the 'Querdenken' movement as well as the general restrictions for demonstrations and public gatherings, limiting the number of participants to a maximum of twenty persons, decreased the momentum of the large, centralized protests focussing on the German legislative located in Berlin. However, the protests did not

stop as a result, but rather changed their character: They became more decentralized and a shift to more informal forms of gatherings and protests could be observed: In several German cities and towns, people gathered for 'Mahnwachen' or 'Spaziergänge' (walks) to protest against the government's COVID-19 politics. Since going for a walk individually was a legal outdoor activity even under the strictest lock-down regulations, this type of action was chosen in order to circumvent the restrictions imposed on demonstrations. In effect, these walks ('Spaziergänge') often resulted in large gatherings of people moving through the streets together, often holding signs or chanting slogans against German politicians or the COVID-19 containment measures in general.

However, the calls to gather for walks in the public became more targeted and mobilized people to gather in front of the private homes of politicians, mostly those in charge for the COVID-19 measures on federal state level.

In December 2021, around thirty members of the far-right group 'Freie Sachsen' (Free Saxonians) gathered in front of the home of Saxony's Minister of Health, Petra Köpping, carrying torchlights. The incident received a lot of public attention, and was criticized in the media as well as by German politicians as attempts to intimidate politicians under the pretext of the protesters right to 'freedom of speech'.

Earlier that year, the same group had gathered in front of Saxony's prime minister, Michael Kretschmer, who is one of the most prominent targets of the radical movement against the COVID-19 containment measures. In December 2021, an investigative report by German journalists revealed that members of the Telegram chat group 'Dresden Offlinevernetzung' had talked about concrete preparations to assassinate Kretschmer. And a few months later, by mid-April 2022, police foiled a plot to kidnap Germany's Minister of Health Karl Lauterbach and to sabotage power facilities in order to create chaos which would ultimately allow to overthrow the government.

The messenger service Telegram played a key role in the mobilization of

the movement. The service allows the creation of large groups and channels to disseminate information quickly to a large number of users. Unlike Facebook or Google, which hand over data to authorities at their request, Telegram does not cooperate with authorities.

This ensures a high degree of anonymity for its users which has proven beneficial for protest movements in authoritarian countries, This anonymity poses a barrier to political repression and criminal prosecution. This makes the service attractive not only for social movements, but also for criminal and terrorist activities, and also facilitates the spread of hate-speech and calls for violence like those mentioned above.

The rise of conspiracy theories in the context of the COVID-19 pandemic has thus not only intensified hate-speech against and the discrimination of certain communities in the form of antisemitism and racism - this process was accompanied by an increasingly toxic and potentially violent climate for politicians in Germany. This makes it once more clear that the digital sphere of social networks and messengers cannot be grasped without also considering its entanglement with and effects for the physical sphere of society. Both online and offline contexts influence and often reinforce each other, as is the case with the mobilization against the COVID-19 containment measures.

The processes illustrated above also point out another important aspect: Even though political activists from Germany's extremist far-right had been involved in the protests from the start, and non-governmental organizations had warned about the antidemocratic and violent potential of parts of the movement, the dangerous potential of this movement seems to have been underestimated by German authorities for a rather long time. Only after journalists had revealed the plans to murder Michael Kretschmer, and after several gatherings in front of private homes of politicians had already taken place, the issue seems to become more of a serious concern.

Thus the process points out the relevance of investigative work by critical journalists and non-governmental organizations to monitor the mobilization

and networking strategies of the far-right spectre in Germany. However, while these actors often have to deal with large amount of data, their investigation often relies on manually conducted research and data evaluation. The potentials of Data Science methods, especially Machine Learning-based approaches for Natural Language Processing (NLP), are often not accessible for these actors, making it hard for them to process large amounts of data from the often very dynamic contexts of informal political networks.

- racism: especially Asian community made responsible for spreading the virus
- antisemitism: Jews identified as culprits behind the pandemic, using it as a means to obtain world dominance
- politicians: complicit with those who benefit from pandemic; following a secret plan; restriction of civil rights
- freedom of speech, criticizing the government are valuable and essential components of a democratic society
- but: often hateful tendencies and developments, both online and offline
- entanglement of digital and real world
- hate does not stay in the internet, also has real effects
- spread of misinformation, conspiracy theories: social media and messengers
- not only information, also mobilization

1.2 Objective

This thesis aims at exploring the potentials of a particular NLP technology, namely Named Entity Recognition (NER) for the investigation of the

mobilization for protests against COVID-19 measures in a German-language Telegram channel. The overarching goal is to explore if a component for Named Entity Recognition could be a beneficial contribution to the Open Source project 'Teledash' (cf. Weichbrodt and Stanjek 2022), to make NER available as a tool for journalists, NGOs, and researchers investigating the COVID-19 related protest movements and networks in Germany.

- approach of NER in German-language Telegram messages
- extraction of entities particularly relevant for mobilization / monitoring of potential targeting: PERSON, ORGANIZATION; DATE; TIME; LOCATION; ACTION as custom type
- explore potentials and limitations of existing models
- explore potentials of translation approach
- explore potentials of transfer learning with Transformer models

1.3 Thesis Structure

- Theory
- Exploration
- Implementation
- Conclusion and Discussion: suggestions for future work

Chapter 2

Theory

The first part of this chapter places Named Entity Recognition in its larger context of Natural Language Processing (NLP) in general, and of Information Extraction (IE) as a specific NLP task in particular. It provides a basic definition of of NER, lines out its historical evolvement, and discusses the challenges related to this task, both in general and with regard to the peculiarities of German language and to text data from social media.

The second part elaborates different approaches for NER, from heuristics-/rule-based methods over classical machine learning-based to deep learning-based approaches including transformers, and briefly discusses the limitations and potentials of the respective approaches.

2.1 Natural Language Processing

The field of Natural Language Processing combines approaches from computer science, linguistics, and artificial intelligence. Its main focus is to build systems and computational techniques for the automated processing and analysis of human language (cf. Vajjala et al. 2020, p. xvii).

Since its evolvement in the 1950s, NLP has been primarily an area of interest for academic research, but lately, it is being used more and more for

real-world tasks and business applications. (cf. Vajjala et al. 2020, p. xvii)

Core NLP applications comprise the use for email platforms (e.g. for spam filters), voice-based assistants, modern search engines, and machine translation services. (cf. Vajjala et al. 2020, p. 5). Furthermore, NLP is used for analyzing web and social media data (e.g. customer reviews), for spelling and grammar correction tools, and for a variety of domains including health care or finance. (cf. Vajjala et al. 2020, pp. 5–6).

Typical NLP tasks are for example language modeling, text classification, text summarization, information extraction, information retrieval, machine translation, or topic modeling (cf. Vajjala et al. 2020, pp. 6–7).

2.2 Information Extraction

Information Extraction (IE) is one of the several NLP tasks mentioned above. The aim of IE is to extract information from unstructured or semistructured natural language texts (cf. Zong, Xia, and Zhang 2021, p. 227).

'Information' can refer to different things, depending on the domain and purpose of the task: The extracted instances can be (named) entities such as people, locations, places, or organizations, but also important events occuring in texts (cf. Vajjala et al. 2020, p. 161). Sometimes the focus is on temporal information such as time and date, or on domain-specific entities, e.g. biomedical instances like genes, proteins, or viruses.

The extracted parts can also be attributes related to those entities, or relationships between the detected entities. The kinds of texts which information is extracted from may be news texts from the web, social media posts, customer reviews, academic literature, and others. The aim of IE is to generate structured output from this unstructured or semistructured text data. Typical tasks of IE are named entity recognition, entity disambiguation, relation extraction, and event extraction (cf. (Zong, Xia, and Zhang 2021, p. 227)

The origins of IE dates back to the late 1970s, and its development was further accelerated in the late 1980s, when the US government financed activities to evaluate IE technology. Organized by the *Defense Advanced Research Projects Agency* (DARPA), the first *Message Understanding Conference* (MUC-1) took place in 1987. The conference provided standard datasets for international researchers to compete on. The MUC evaluation was held seven times between 1987 and 1997. The tasks included NER, coreference resolution, relation extraction, and slot filling, and mainly focused on text data from limited domains such as naval military intelligence, terrorist attacks, or aircraft crashes (cf. Zong, Xia, and Zhang 2021, p. 228). The first NER task was introduced at the Sixth Message Understanding Conference (MUC-6) in 1996. The aim was to apply information extraction technologies for largely domain independent named entity recognition. Besides recognizing the entity types person, organization, and location, the task also involved numerical expressions including time, currency, and percentage (cf. Grishman and Sundheim 1996). The corpus for the task consisted of 318 annotated articles from the Wall Street Journal.

By the end of the 1990s, the MUC was replaced by the *Automatic Content Extraction* (ACE) meeting which put its focus on extracting more fine-grained entities, as well as entity relations, and events. There was also a shift in the data to broader news data covering political and international events.

In 2009, ACE became a subtask of the *Text Analysis Conference Knowledge Base Population* (TAC KBP). This shifted the focus in data to open domain data (mainly from webpages), and the focus of tasks to entity attribute extraction and entity linking. Both MUC and ACE contributed to the development of several standard test datasets for the evaluation and development of IE approaches. (cf. Zong, Xia, and Zhang 2021, p. 229).

Following Zong, Xia, and Zhang 2021, IE technology can be classified via the domain aspect (limited vs. open domain), the type of extracted information (entity recognition, relation extraction, event extraction), and implemen-

tation methods (rule-based, machine learning-based, or deep learning-based approaches) (cf. Zong, Xia, and Zhang 2021, p. 229).

Common examples of IE applications are news tracking, counter-terrorism, business and financial intelligence, the processing of biomedical data or scientific libraries. IE can also be utilized for applications building on entity-oriented search, for question-answering systems, or the task of text segmentation (cf. Aggarwal 2018).

2.3 Named Entity Recognition

Named entity recognition is a subtask of IE and addresses the task of identifying named entities like person, location, organization, drug, time, clinical procedure, biological protein, etc. in text. NER is often used as the first step in question answering, information retrieval, co-reference resolution, topic modeling, etc. and is thus a fundamental task in NLP (cf. Yadav and Bethard 2019).

NER can be divided into the following two subtasks (cf.Zong, Xia, and Zhang 2021, p. 229):

- **1. Entity detection**, to detect whether a given string in a text document is an entity, including the detection of the entity's boundaries.
- **2. Entity classification**, to assign a specific category to the detected entity.

2.3.1 Common entity types

The entities which are to be identified via NER approaches are commonly represented by the following categories:

- PERSON
- LOCATION

- ORGANIZATION
- MISC/OTHER

Early NER approaches also included time, date, currency/quantity, and percentage as entity types. But while these can commonly be identified via regular expressions, instances of persons, locations, organizations, and miscellaneous/other usually do not follow a consistent pattern and thus pose larger challenges for NER, which is why most of NER research focusses mainly on these entity types (cf. Zong, Xia, and Zhang 2021, p. 229).

Also it should be noted that times, dates, currencies, quantities and percentage do not qualify as named entities in a narrow sense, but may be useful depending on the purpose of the application which NER is used for (cf. Aggarwal 2018, p. 386).

2.3.2 IOB tagging scheme

Since NER is usually regarded as a sequence labeling task, an appropriate label set as well as the language granularity for the annotation process needs to be defined for the respective sequence labeling model.

A widely used tagging scheme is the IOB (or BIO) label set. In this tagging scheme, the B-tag is used to label the beginning of an entity, the I-tag is used for labeling intermediate or final tokens of an entity, and the O-tag is assigned to all tokens outside of an entity.(cf. Zong, Xia, and Zhang 2021, p. 231).

Instances of different entity types representing persons, locations, or organizationa are thus labeled with two different tags per entity type: B-PER, I-PER for persons, B-LOC, I-LOC for locations, and B-ORG, I-ORG for organizations (cf. Zong, Xia, and Zhang 2021, pp. 231–232). The usual granularity levels for the annotation of entities are usually word- and character-level (cf. Zong, Xia, and Zhang 2021, p. 232).

2.3.3 Evaluation of NER systems

Similar to other NLP tasks, NER involves an evaluation of the trained model. Evaluation can be done during model building, deployment, as well as in production, and rely on different metrics depending on the task at hand. Usually, the evaluation focuses on the model's performance on unseen data.

The evaluation during model building and deployment usually relies on intrinsic evaluation using ML metrics, while the evaluation in the production phase also involves application-specific business metrics, which is called extrinsic evaluation.

Extrinsic evaluation is usually way more expensive than intrinsic evaluation, and often requires the involvement of stakeholders, sometimes even end-users. **Intrinsic evaluation** using ML metrics on the other hand can be done by machine learning engineers themselves and is less cost-effective. Also, they are applied in the earlier phase of the process, thus allowing early adjustments and quick improvments of the model (cf. Vajjala et al. 2020, p. 71). This is why intrinsic evaluation is usually used as a 'proxy for extrinsic evaluation.' ([71]Vajjala et al. 2020).

To measure a model's performance, usually a test set is selected before training the model, and excluded from the training process. After training the model on the training set, the test set is used to evaluate the predictions of the model to assess its performance on unseen data. For NER, the manually annotated entity tags in the test set serve as gold reference with which the models predictions are compared (cf. Zong, Xia, and Zhang 2021, p. 241).

While intrinsic evaluation is a standard measure on most ML tasks, the specific metrics used differ depending on the specific task.

F1, Precision, and Recall

For NER models, the commonly used metrics are **precision**, **recall**, and **F1-score**.

These metrics rely on the number of entities recognized correctly by the

model (true positives, TP), the number of entities detected by the model, but not labeled as entities in the test set (false positives, FP), and the number of entities not recognized by the model, but tagged as named entities in the test set (false negatives, FN).

Precision, recall, and F1 are defined as follows:

$$precision = \frac{TP}{TP + FP} * 100$$

$$recall = \frac{TP}{TP + FN} * 100$$

$$F1 = \frac{2 * precision * recall}{precision + recall}$$

Precision represents the percentage of correctly identified named entities out of all recognized named entities, (i.e. the percentage of true positives out of all positive predictions), and recall equates to the percentage of named entities that are recognized by the model out of all tokens tagged as named entities in the test set.

The F1-score combines precision and recall and is the metric commonly used in shared tasks on NER to assess the performance of the participating systems.

2.3.4 General Challenges for NER

Since NLP deals with human language, it is confronted with several challenges which stem from the characteristics of human language:

Ambiguity, understood as uncertainty or inexactness of meaning, is inherent to most human languages: Words or phrases can have multiple meanings, and usually it requires knowledge of the context in which they appear to resolve this ambiguity. Making computer-based systems take into account and 'understand' these contexts is one of the major challenges for NLP (cf.

Vajjala et al. 2020, pp. 12–13) As for NLP tasks in general, ambiguity is also a core challenge for NER (cf. Aggarwal 2018, p. 386). For example, the word 'Bill' might refer to a persons first name and thus indicate a named entity of type PERSON, but when located at the beginning of the sentence, it might also be a synonym for 'invoice'.

For resolving this ambiguity, the **context** of a given word is usually of crucial relevance (cf. Aggarwal 2018, p. 387). NER systems differ in terms of how they can or cannot take into account the context of a word. Moreover, the data itself might not offer enough context. While articles from newspapers usually are of sufficient length and contain sufficient contextual information, social media texts are often of shorter length which results in a lack of context.

Common knowledge is often an essential part in a conversation between humans: Facts are assumed to be known and thus not expressed explicitly in statements. Humans constantly use common knowledge to understand language. One of the challenges for NLP is thus how to encode this common knowledge for computational approaches (cf. Vajjala et al. 2020, pp. 13–14).

Creativity is another feature inherent to human language: Even though language is constituted via and essentially shaped by rules, it is far from being exclusively rule-driven. Rather, language is creative and heterogenuous; characterized by various styles, genres, and dialects. One important challenge for NLP is thus to enable machines to adequately handle this creativity (cf. Vajjala et al. 2020, p. 14). Moreover, language changes and evolves over time, in interaction with social and cultural processes, as well as with the development of communication technology (one example being the emergence of hashtags in the context of social media). NLP applications developed using data from the past might thus not fit well for data consisting of newly evolving manifestations of human language. These creative and dynamic characteristics of human language also pose challenges for the task of named entity recognition: Since new entities evolve over time, the set of

know entities is never constant. While for 'some types of entities like locations, the names of entities evolve slowly, whereas for others like people and organizations, the incompleteness problem is very significant.' (Aggarwal 2018, p. 386).

Abbreviations referring to multiple instances of the same named entity are another important challenge (cf.Aggarwal 2018, p. 387). For example, the terms 'U.S.', 'USA' or 'United States of America' all refer to the same geopolitical entity. To overcome this challenge, usually entity disambiguation is an important step following named entity recognition.

Yet another challenge stems from **capitalization** conventions which vary across languages: While in English-language texts, capitalization usually indicates proper names and can thus be used as a feature for named entity recognition, in German-language texts all nouns are capitalized, making the feature less suitable (cf. Benikova, Biemann, and Reznicek 2014).

Diversity across languages makes it hard to apply an NLP solution from one language to another. According to the database *Ethnologue*, there are currently over 7.000 languages actively spoken worldwide (cf. Ethnologue 2022). Even though some of these languages share the same family and thus have some similarities, a direct mapping between two languages is not possible. Building language-agnostic solutions is a very complex matter, while building separate solutions for each language is labor- and time-intensive (cf. Vajjala et al. 2020, p. 14).

Imbalanced resources for different languages is yet another challenge for NLP, both in terms of research efforts and data. In fact, most NLP technologies are still designed for English or other European languages (cf. Singh 2008). For these so-called high-resource languages, a large number of datasets and libraries exists while low-resource languages are usually significantly 'less studied, resource scarce, less computerized' (Magueresse, Carles, and Heetderks 2020). This imbalance is also reflected in the existing NER datasets.

2.3.5 Specific Challenges for NER in User-Generated Texts

The aforementioned challenges for NER also apply for texts from social media platforms. However, named entity recognition for these user-generated texts can be regarded as even more challenging.

One reason for this is the challenge of **domain adaptation**: Most of the existing NER systems are trained on a handful of standard NER datasets which mainly consist of newswire texts. Since social media text data is often **more informal and noisy** (e.g. due to missspellings) than newspaper articles, which aggravates domain adaptation for this genre and usually results in poor performance of existing NER systems when applied to texts from social media (cf. Ritter et al. 2011).

Named entity recognition applications for other domains thus usually require the creation of custom corpora including a labor-intensive annotation process.

Furthermore, since texts from social media are often **shorter of length**, they lack sufficient context for the determination of an entity's type (cf. Ritter et al. 2011).

The **wide variety of capitalization styles** are another challenge, like e.g. the use of only lowercase for German texts, or the use of uppercase only in some texts (cf. Ritter et al. 2011). As Ritter et al. 2011 points out, news-trained NER systems seem to rely heavily on capitalization which limits their applicability to user-generated social media texts.

Figure 2.1 illustrates the noisy character of tweets as an example.

1	The Hobbit has FINALLY started filming! I cannot wait!
2	Yess! Yess! Its official Nintendo announced today that they Will release the Nintendo 3DS in north America march 27 for $250
3	Government confirms blast n nuclear plants n japan...don't knw wht s gona happen nw...

Figure 2.1: Examples of noisy text in tweets, taken from Ritter et al. 2011

Also, these texts usually contain more **out of vocabulary (OOV) words** than e.g. news articles. This is partly due to misspellings, but also stems from platform specific codes and styles: The frequent occurence of **special characters** such as # for hashtags or for user mentions poses challenges because they cannot easily be handled during annotation and training. User-generated text in social media also often includes a lot of **lexical variations**. As an example, Ritter et al. 2011 collected more than fifty variations for the word 'tomorrow' from a Twitter corpus, as shown in figure 2.2.

‘2m’, ‘2ma’, ‘2mar’, ‘2mara’, ‘2maro’, ‘2marrow’, ‘2mor’, ‘2mora’, ‘2moro’, ‘2morow’, ‘2morr’, ‘2morro’, ‘2morrow’, ‘2moz’, ‘2mr’, ‘2mro’, ‘2mrrw’, ‘2mrw’, ‘2mw’, ‘tmmrw’, ‘tmo’, ‘tmoro’, ‘tmorrow’, ‘tmoz’, ‘tmr’, ‘tmro’, ‘tmrow’, ‘tmrrow’, ‘tmrrw’, ‘tmrw’, ‘tmrww’, ‘tmw’, ‘tomaro’, ‘tomarow’, ‘tomarro’, ‘tomarrow’, ‘tomm’, ‘tommarow’, ‘tommarrow’, ‘tommoro’, ‘tommorow’, ‘tommorrow’, ‘tommorw’, ‘tommrow’, ‘tomo’, ‘tomolo’, ‘tomoro’, ‘tomorow’, ‘tomorro’, ‘tomorrw’, ‘tomoz’, ‘tomrw’, ‘tomz’

Figure 2.2: Lexical variations of the word ‘tomorrow’ in tweets, taken from Ritter et al. 2011

Last but not least, social media texts often **mixed-language texts**, making it harder for language-specific models to accurately determine named entities in them.

2.3.6 Important Shared Tasks and Datasets for German-Language NER

The NER tasks at the Conference on Computational Natural Language Learning (ConLL) for English and German in 2003 was an important cornerstone for German-language NER (cf. Tjong Kim Sang and De Meulder 2003)

The provided German dataset consists of 909 articles sampled from the newspaper Frankfurter Rundschau (cf. Tjong Kim Sang and De Meulder

2003). The task focused on the entity types person (PER), location (LOC), organization (ORG), and miscellaneous (MISC).

The ConLL 2003 dataset is publicly accessible, but no licensing information is available. Furthermore, given the fact that the corpus only contains newspaper articles, its applicability to other domains might be limited. The annotation of the dataset was carried out at the University of Antwerp (cf. Tjong Kim Sang and De Meulder 2003), thus most likely by non-native speakers, which might result in inconsistencies of the labeled data.

In 2014, GermEval, a series of shared tasks focussing on Natural Language Processing for German-language, announced a task for Named Entity Recognition for German-language texts (cf. Benikova, Biemann, and Reznicek 2014).

The provided dataset was compiled from German Wikipedia articles and online news. Similar to the ConLL 2003 task, entity types of interest were person (PER), organization (ORG), location (LOC), and other (OTH).

The dataset is publicly available under a e CC-BY license. However, since the GermEval2014 dataset also includes nested and derived entities, it is not readily combinable with other datasets. And even though the corpus was created from online resources, it does not contain noisy user-generated texts, limiting its applicability to other social media texts.

Up until today, the Germeval 2014 corpus and (to some extent) the ConLL 2003 dataset represent standards for German-language NER and are used to train and evaluate state-of-the art NER models.

2.3.7 Important Shared Tasks, Experimental Studies, and Datasets for NER on Social Media Texts

In 2011, Ritter et al. conducted an experimental study on named entity recognition in tweets, and used a random sample of 2.400 English-language tweets for this which were annotated with the following ten entity types: PERSON, GEO-LOCATION, COMPANY, PRODUCT, FACILITY, TV-SHOW,

MOVIE, SPORTSTEAM, BAND, and OTHER (cf. Ritter et al. 2011). The best participating system in this task achieved an F1 score of 0.66 for these ten types (cf. Ritter et al. 2011).

A more detailed look into the results for the standard entity types PERSON, LOCATION, and ORGANIZATION, reveals that the F1 score for the entity type PERSON ranks highest with up to 0.83, followed by LOCATION, and then ORGANIZATION (0.74 and 0.66) for the best participating system.

In 2015, Derczynski et al. investigated the applicability of existing, by then state-of-the-art systems for named entity recognition in tweets. Their results indicate as well that existing machine-learning based approaches do not perform well on noisy user-generated data from microblogs, using three different English-language datasets, including the corpus created by Ritter et al. 2011.

The evaluated systems achieved F1 scores ranging between 40.27, 51.49 (Ritter dataset), and 77.18 (cf. Derczynski et al. 2015).

In line with Ritter et al. 2011, the authors identify unreliable capitalization, typographic errors, shortenings, and lack of context due to the shortness of the texts as main challenges in this specific domain.

The authors suggest language detection, part-of-speech tagging, and normalization as pre-processing steps to encounter these challenges. Their results indicate however that e.g. normalization only leads to minimal increases, and in some cases even decreases the F1 score (cf.Derczynski et al. 2015).

Also in 2015, Baldwin et al. organized a shared task for Named Entity Recognition on social media data, namely tweets from Twitter in the context of the *Workshop on Noisy User-generated Text* (W-NUT). The task used the corpus created by Ritter et al. 2011.

Among the participating systems, the best F1 score achieved was 56.41, which is ∼0.1 lower than the F1 score achieved in the initial study by Ritter et al. 2011, and ∼0.2 lower than the best F1 score achieved at the GermEval

2014 NER task (76.38, cf. Benikova et al. 2014).

Unfortunately: no german-language WNUT tasks

2.3.8 Additional German-language datasets for NER

NER German-language datasets already mentioned: ConLL 2003, GermEval 2014; are used for training and for evaluation up until today ('gold standard')

additionally: Smartdata corpus, XTREME corpus, WIKIANN?

2.4 Overview of NER approaches

The following section provides an overview of NER approaches as they have evolved over time: Starting with early rule-based and knowledge-based systems, the field moved on to machine-learning approaches based on feature-engineering. The development of NER systems based on neural networks with minimal feature engineering mark another important step forward in the field. Lately, the introduction of transformer models brought along yet another crucial advancement and achieve new state of the art performance for several NLP tasks including NER.

2.4.1 Rule-based and knowledge-based approaches

The first systems for NER were relied on handcrafted rules, lexicons, orthographic features and ontologies (cf. Yadav and Bethard 2019).

Rule-based systems make use of the fact that the structure and the context of named instances of type person, location, and organization often follow certain rules and can thus be recognized e.g. by using regular expressions (cf. Zong, Xia, and Zhang 2021, p. 230). Hand-crafted rules can make use e.g. of capitalization (used in a lot of languages for the beginning of person names) or of certain 'key words' such as titles for persons (Mr., Mrs.

Dr., etc.) for the recognition of named entities (cf. Zong, Xia, and Zhang 2021, p. 230).

However, these approaches are limited even when a large amount of data is considered. This is due to the ambiguity of language, resulting in different meanings for a given word or sequence in different contexts. Also, these approaches are rather static and often struggle with handling newly emerging identities without updating the underlying rules. Furthermore, the generation of rules is labor-intensive and usually requires expert domain knowledge. (cf. Zong, Xia, and Zhang 2021, p. 231).

Knowledge-based approaches rely on lexicon resources and domain specific knowledge. The performance of these systems heavily depends on the exhaustiveness and completeness of the used lexicon. Here as well, the process can be rather labor-intensive because domain expert knowledge is needed for the construction and maintenance of the knowledge resources (cf.Yadav and Bethard 2019).

2.4.2 Machine-learning based approaches

Machine-learning based approaches based on feature-engineering introduced a critical advancement for the task of NER (cf. Nadeau and Sekine 2007)

Machine learning "deals with the development of algorithms that can learn to perform tasks automatically based on a large number of examples, without requiring handcrafted rules." (cf. Vajjala et al. 2020, p. 14)

This allows for developing NER models through an automated learning process.

goal of ML: 'learn' to perform tasks based on examples (training data) without explicit instruction typically done by creating a numeric representation (features) of the training data and using this representation to learn the patterns in those examples (cf. Vajjala et al. 2020, p. 15)

ML algorithms can be grouped in three paradigms: supervised learning, unsupervised learning, reinforcement learning

supervised: learn the mapping function from input to output given a large number of examples in the form of input-output pairs input-output pairs = training data output = labels, or ground truth example: spam classification

supervised learning requires large number of labeled data (cf. Vajjala et al. 2020, p. 15)

Feature-engineered supervised systems

Given a collection of text data, suppose all person, location, and organization entities are manually labeled, as shown in the examples below. Supervised NER systems attempt to design machine learning methods that learn automatic prediction models based on these correctly labeled training data.Zong, Xia, and Zhang 2021, p. 231

- Researchers usually regard this task of supervised NER as a sequence labeling problem.

Supervised machine learning models learn to make predictions by training on example inputs and their expected outputs, and can be used to replace human curated rules. Common ML systems for NER: Hidden Markov Models (HMM) Maximum Entropy Support Vector Machines (SVM) Conditional Random Fields (CRF) decision trees Yadav and Bethard 2019

Support Vector Machines - goal of any classification task: learn a decision boundary that acts as a separation between different categories of text - decision boundary can be linear or nonlinear (e.g. a circle) - SVM: can learn linear and nonlinear - SVM learns an optimal decision boundary so that the distance between points across classes is at its maximum Vajjala et al. 2020, p. 20

Hidden Markov Model HMM is a statistical model that assumes there is an underlying, observable process with hidden states that generates the data, ie we can only observe the data once it is generated - an HMM then

tries to model the hidden states from this data - e.g. POS tagging: we assume that text is generated according to an underlying grammar, which is hidden underneath the text - hidden states are parts of speech that inherently define the structure of the sentence following the language grammar, but we only observe the words that are governed by these latent states - HMM also make the Markow assumption: each hidden state is dependent on the previous state(s) - human language is sequential which means that the current word depends on what occured before - this makes HMM a powerful tool for modeling textual data Vajjala et al. 2020, pp. 21–22

Conditional Random Field - another algorithm used for sequential data - performs a classification task on each element in the sequence - takes the sequential input and the context of tags into consideration Vajjala et al. 2020, p. 22

Unsupervised / bootstrapped approaches

Unsupervised learning: aim to find hidden patterns in given input data without any reference to output works with large collection of unlabeled data example: topic modeling, identity latent topics in large collection of texts (cf. Vajjala et al. 2020, p. 15)

common in real-world NLP: semi-supervised; small labeled dataset, large unlabeled dataset; learn from both sets (cf. Vajjala et al. 2020, p. 16)

reinforcement learning: absence of larger data colletions, trial and error principle, improvement via feedback (punishment or reward) (cf. Vajjala et al. 2020, p. 16)

Some of the earliest systems required very minimal training data. - labeled seeds, and 7 features including orthography (e.g., capitalization), context of the entity, words contained within named entities, etc. - shallow syntactic knowledge and inverse document frequency (IDF) - uses seeds to discover text having potential named entities, detects noun phrases and fil-

ters any with low IDF values, and feeds the filtered list to a classifier Yadav and Bethard 2019

2.4.3 The introduction of neural networks

Deep learning: "branch of machine learning that is based on artificial neural network architectures." (cf. Vajjala et al. 2020, p. 14)

NER systems have been studied and developed widely for decades but accurate systems using deep neural networks (NN) have only been introduced in the last few years (Yadav and Bethard 2019)

Starting with Collobert et al. (2011), neural network NER systems with minimal feature engineering have become popular

are appealing because they typically do not require domain specific resources like lexicons or ontologies, and are thus poised to be more domain independent. Various neural architectures have been proposed, mostly based on some form of recurrent neural networks (RNN) over characters, sub-words and/or word embeddings. Yadav and Bethard 2019

Feature-inferring neural network systems Collobert and Weston (2008) proposed one of the first neural network architectures for NER, with feature vectors constructed from orthographic features (e.g., capitalization of the first character), dictionaries and lexicons.Yadav and Bethard 2019

Later work replaced these manually constructed feature vectors with word embeddings (Collobert et al., 2011), which are representations of words in n-dimensional space, typically learned over large collections of unlabeled data through an unsupervised process such as the skip-gram model (Mikolov et al., 2013). Studies have shown the importance of such pre-trained word embeddings for neural network based NER systems (Habibi et al., 2017), and similarly for pre-trained character embeddings in character-based languages like Chinese Modern neural architectures for NER can be broadly classified into categories depending upon their representation of the words in a sentence. For example, representations may be based on words, characters,

other sub-word units or any combination of these Yadav and Bethard 2019

Feedforward neural networks

2.4.4 Recurrent neural networks

Recurrent neural networks - language is inherently sequential - thus, a model which can progressively read an input text from one end to another can be very useful for language understanding - RNNs are designed to keep such a sequential processing and learning in mind - have neural units that are capable of remembering what they have processed so far - memory is temporal, information is stored and updated with every step as the RNN reads the next word as input Vajjala et al. 2020, p. 23

Long short-term memory - RNNs cannot remember longer contexts - do not perform well when input text is long - LSTM = solve this issue - concept: forget irrelevant context, only remember relevant context for task at hand - relieves the load of having to remember a lot of context for long inputsVajjala et al. 2020, p. 23

Bi-LSTM

Convolutional neural networks - used heavily in computer vision tasks (e.g. image classification) - also used for NLP - each word in a sentence is represented as a word vector, all word vectors of sentence have the same size - resulting matrix representing the sentence can now be treated similar to an image by the CNN - advantage: CNN can look at a group of words together using a context window - CNN uses a collection of convolution and pooling layers to achieve a condensed representation of the text which is then fed as input to a fully connected layer to learn some NLP tasks like text classification Vajjala et al. 2020, p. 24

2.4.5 Transformers

- latest entry in the league of deep learning models for NLP - have achieved state of the art in almost all major NLP tasks in recent years - model the textual context but not in a sequential manner - given a word in the input, it prefers to look at all the words around it (known as self-attention) and represent each word with respect to its context - e.g. bank different meaning depending on context (financial institute, furniture) - transformers can model such context and hence have been used heavily in NLP tasks due to this higher representation capacity as compared to other deep networks Vajjala et al. 2020, pp. 25–26

- prior to transformers: recurrent architectures (e.g. LSTMs, RNNs) - recurrent architectures contain a feedback loop in the network connections; allows information to propagate from one step o another, making them ideal for modeling sequential data like text - RNN: receives input, feeds it through the network, outputs a vector called the hidden state - at the same time, the model feeds some information back to itself through the feedback loop, which it can then use in the next step - RNN passes information about its state at each step to the next operation in the sequence - this allows an RNN to keep track of information from previous steps, and use it for its output predictions Tunstall et al. 2022, p. 2

Transfer learning for downstream tasks with transformers - large transformers used for smaller downstream tasks via transfer learning - transfer learning = technique where the knowledge gained while solving one problem is applied to a different but related problem - transformers: pre-training (training of very large transformer models, unsupervised) to predict a part of a sentence given the rest of the content so it can encode the high-level nuances of the language in it - models are trained on more than 40GB of textual data, scraped from the whole internet - this model is then fine-tuned on NLP downstream tasks, e.g. NER Vajjala et al. 2020, p. 26

2.5 Summarization

Chapter 3

Exploration of existing solutions

In order to explore the potentials and limitations of existing German-language models for Named Entity Recognition, two models were selected for a qualitative exploration of their results with some sample texts.

Furthermore, it was explored if machine translation could be used as a tool to allow for the use of well-resourced English NER models for the recognition of named entities in (originally German-language) texts after translating them to English.

3.0.1 Models

The two models explored in this chapter, `spaCy` and `Flair`, were chosen due to their easy applicability, allowing for fast exploration and analysis, and because both of them support both German- and English-language named entity recognition.

spaCy is a free open-source library for Natural Language Processing developed for use in production. The library is extensively documented and thus serves as an easy starting point for beginner, while also claiming to be de-

signed specifically for use in production. The library supports more than 66 languages (including German), comes with pre-trained word vectors, and supports a variety of NLP tasks such as part-of-speech tagging, text classification, and named entity recognition.

While the English `spaCy` model supports named entity recognition for 18 entity types (`CARDINAL`, `DATE`, `EVENT`, `FAC`, `GPE`, `LANGUAGE`, `LAW`, `LOC`, `MONEY`, `NORP`, `ORDINAL`, `ORG`, `PERCENT`, `PERSON`, `PRODUCT`, `QUANTITY`, `TIME`, and `WORK_OF_ART`), the German `spaCy` model is unfortunately limited to the common entity types are `LOC`, `MISC`, `ORG`, and `PER`.

Schweter and Akbik 2020

3.0.2 Data

For the exploration of these models, a few example messages were randomly selected from the Telegram channel 'Demotermine' (*Demo dates*).

The channel was selected because a prior qualitative exploration of sample messages indicated that its content consists mainly of information about and mobilization for demonstrations and others forms of protests against the German COVID-19 containment measures and could thus be expected to contain a lot of relevant entities.

3.1 German-language Named Entity Recognition with spaCy and Flair

3.1.1 Implementation

For exploring the German s`spaCy` model, a simple code was implemented, using the `displacy` module of the library to display the sample texts tagged with the detected entities.

For exploring the German medium-size spaCy model, a simple code was

implemented, using the displacy module of the library to display the sample texts tagged with the detected entities.

```
import spacy
from spacy import displacy

def display_entities(texts):
  for text in texts:
    try:
      doc = nlp(text)
      sentence_spans = list(doc.sents)
      displacy.render(sentence_spans, style="ent")

        except:
          pass

nlp = spacy.load("de_core_news_md")
display_entities(samples)
```

The next code section shows the implementation for the Flair model:

```
from flair.data import Sentence
from flair.models import SequenceTagger

tagger_default = SequenceTagger.load("flair/ner-german")

def get_ents(tagger, text):
  sentence = Sentence(text) # predict NER tags
  tagger.predict(sentence)
  # print sentence
  print(sentence)
  # print predicted NER spans
  print('The following NER tags are found:')
  # iterate over entities and print
  for entity in sentence.get_spans('ner'):
    print(entity)

for text in samples:
```

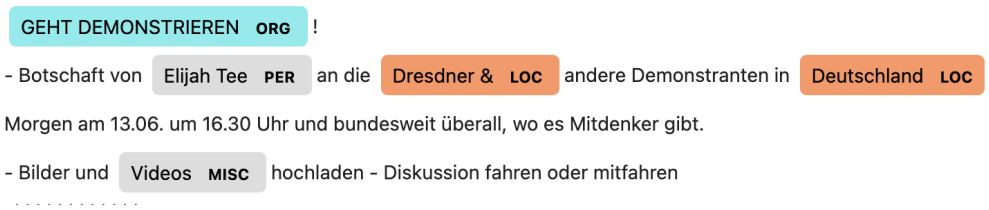
Table 3.1: Detected entities for German-language example messages

No.	Text	spaCy	Flair
1	Freiburg Aufzug mit Endkundgebung am Platz der alten Synagoge . Live Musik ! Details 19062021 Raus auf die Straße	Endkundgebung[LOC], alten Synagoge[LOC]	Freiburg[LOC]

```
try:
  get_ents(tagger_default, text)
except:
  pass
```

3.1.2 Results

The results show that the model does to some extent recognized locations and persons mentioned in the example texts:



However, the assignment of the labels `ORG` to the phrase 'GEHT DEMONSTRIEREN' and `MISC` to 'Videos' is irritating. Also, the text part 'Dresdner &' was labeled as location, which is wrong since the text here refers residents of Dresden, but not to the city Dresden itself.

The next example leads to strong doubts about the model's capability to recognized locations in a text: While 'Blankenfelde' is correctly labeled, the model misses the remaining three locations mentioned in the text, 'Mahlow',

'Berlin', and 'Lichtenrade'. Furthermore, the labeling of the phrase 'Out on the road' as `MISC` is not plausible. Nor is the classification of the phrase 'Der Mann mit dem Schild' (the man with the sign) as `MISC` in the next example text understandable:

Admittedly, these explorations are rather anecdotal and cannot be generalized without further evaluation based on quantiative metrics. However, the claim that `spaCy` offers production-ready components is put into question when looking at these examples.

3.2 The German-language Flair model

better performance, but still: problem: time, date, action cannot be recognized since not learned by model

3.3 Exploring the multilingual XLM Roberta large finetuned for NER

model with most downloads for German / multilingual NER disadvantage: large -¿ requires more resources exploration... problem: time, date, action cannot be recognized since not learned by model

all models trained on Wikipedia or GermEval -¿ not social media data rather few datasets for German NER, almost all models are evaluated on GermEval14 or even CoNLL 2003!

3.4 Potentials of Machine Translation for NER

Machine translation: sometimes used in the context of augmentation (create training data for multilingual models) some attempts to use MT for low-resource languages

small experiments: translated some example texts with DeeplAI, then fed to spacy and flair

spacy English: knows more entity types flair: more training data because of English

3.5 Summarization

potentials and limitations / observed errors - spacy - Flair - XLM Roberta

explorative evaluation

Chapter 4

Implementation

4.1 Data

4.1.1 Existing German-language datasets for this specific task

GermEval 2014 NER task dataset

Why not GermEval? TIME, DATE missing; ACTION as well not social media data

DFKI Smartdata corpus

the DFKI SmartData Corpus is released as CC-BY 4.0.

XTREME dataset

PAN-X annotations were generated through an automated process -¿ only 'silver standard', not used here

WIKIANN

?

4.1.2 Domain-specific data from Telegram channel

For the creation of a dataset suitable for the aim of this thesis, the Telegram channel 'Demotermine' (*Demo dates*) was selected.

The channel was selected because a prior qualitative exploration of sample messages indicated that its content consists mainly of information about and mobilization for demonstrations and others forms of protests against the German COVID-19 containment measures and could thus be expected to contain a lot of entities of the types relevant for the aim of this thesis. The focus of the channel is also reflected in the extended channel name 'Demo-Termine & Kontakte, Gleichgesinnte treffen - Demokalender, Spaziergänge, Mahnwachen, Meditation *Rücktritt Bundesregierung!' (*Demo dates & contacts, meet like-minded - Demo calendar, walks, vigils, meditation *resignation federal government!*).

The channel was part of a network investigated by IDZ Jena, and in the context of the research project 'Digitaler Hass', data from the channel was collected including messages from the start of the channel until December 19, 2021.

By the beginning of June 2022, the channel had around 52.900 subscribers.

The dates of the collected messages range between March 3rd, 2018 and December 19th, 2021 (last day included in the data collection process). Even though the channel existed before the outbreak of the COVID-19 pandemic, an exploration of message frequencies clearly reveals that the channel gained relevant momentum at the beginning of the pandemic in Germany in spring 2020, as can be seen in figure 4.1.

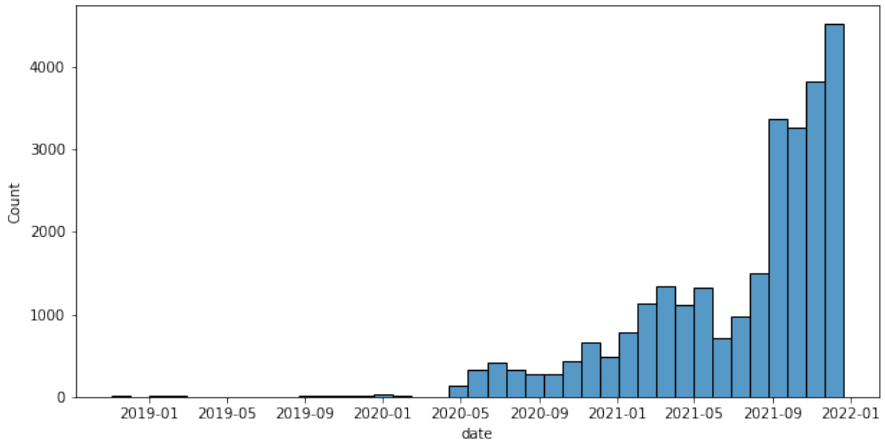


Figure 4.1: Message frequencies

4.1.3 Text cleaning

The initial data collection yielded 31.430 rows of data. After the removal of text duplicates and rows not containing any text, 27.627 rows remained for further processing.

`@user_mentions` or `#hashtags` were deliberately included, and the special characters and words of these tokens were not split up into separate tokens. In-word hyphens (such as in Demo-Termine) were also not split up from their surrounding words. For this, the tokenizer of the German medium-size `Spacy` model `de\_core\_news\_md` was applied with a customized token match pattern as shown in the following code snippet:

```
import re
import spacy
from spacy.tokenizer import _get_regex_pattern

model="de_core_news_md"
nlp = spacy.load(model)
```

```
# code to keep special tokens as a sequence, taken from:
                                https://stackoverflow.com/
                                questions/43388476/how-could-
                                spacy-tokenize-hashtag-as-a-
                                whole
# get default pattern for tokens that don't get split
re_token_match = _get_regex_pattern(nlp.Defaults.token_match)
# add patterns (here: #hashtags, in-word hyphens, @mentions)
re_token_match = f"({re_token_match}|#\w+|\w+-\w+|@\w+)"
# overwrite token_match function of the tokenizer
nlp.tokenizer.token_match = re.compile(re_token_match).match
```

Preventing the special characters in hashtags and user mentions from being split up was intended to serve for creating a realistic social media corpus and allow to include these special characters into the labels assigned during the annotation process. However, evaluation of training results indicate that it might indeed make sense to split up these special characters and simply treat them as separate tokens which are not treated as part of the tagged named entities.

The overall text cleaning process was restricted to an absolute minimum in order to create a sample data which is as realistic as possible with regard to the task of recognizing named entities in social media text. This basically included the removal of newlines and splitting up of compound hashtags into separate hashtags (so that e.g. #food#delicious would be split up into two different tokens #food and #delicious). The following code demonstrates the respective steps, including the application of the customized tokenizer.

```
import re
import spacy
from spacy.tokenizer import _get_regex_pattern

def preprocess_text(s, nlp):
    # remove newlines
    s = ' '.join(s.split())
```

```
    # split up compound hashtags, e.g. #food#delicious
    s = re.sub(r'#', r' #', s)
    doc = nlp(s)
    # tokenize keeping all tokens, no removal of emojis
    words = [token.text for token in doc]
    # join tokens back together
    result = " ".join(words)
    #replace duplicate whitespaces
    result = re.sub(r'  ', '', result)
    return result
```

When applying the `spacy` tokenizer, not only alphabetic tokens but all tokens were obtained. This was due to the fact that tokens representing dates and times usually consist of numbers and punctuation. Thus, in order to include dates and times in the sample data, all tokens were retained. URLs and emojis were also included in the final data, and the capitalization of tokens was retained as well.

The above code parts were then applied to all texts in the dataset, yielding a new column including all the cleaned text values:

```
from datetime import datetime

ts_before = datetime.now()
df['cleaned_text'] = df['text'].apply(lambda x:
                                      preprocess_text(x, nlp) if len
                                      (x)> 0 else None)
ts_after = datetime.now()
print(f"Took {ts_after - ts_before}")
print(len(df), "rows remaining.")
```

```
Output:
Took 0:03:43.779192
27627 rows remaining.
```

After this step, rows not containing values for cleaned text were removed from the dataset, leaving 22.803 messages for sample selection.

4.1.4 Sample Selection

For sample selection, first the text length for each text was obtained to enable filtering texts based on their length. The second step consisted of detecting the language for each text in the dataset in order to be able to filter out non-German texts. For this, the `Python` package `langdetect` was applied to the cleaned texts.

After performing language detection on all messages in the channel, results show that German-language content is clearly predominant with $\sim$ 83% (22.803 messages), followed by English ($\sim$ 3.8%, 3.823 messages), and then French, Italian, and Dutch (¡1% each).

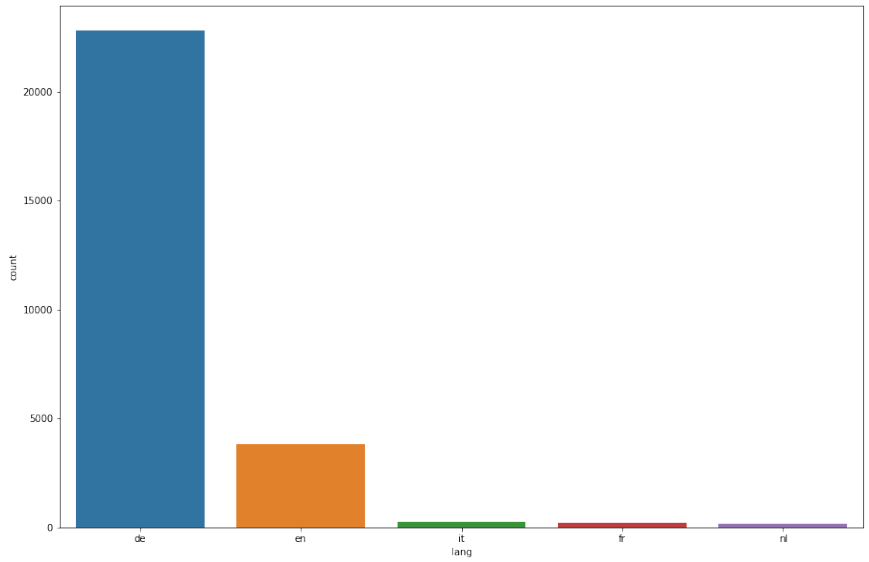


Figure 4.2: Five most frequent languages

It should be noted that the language detection was sometimes flawed due to the fact that the channel contains a number of mixed-language texts resulting in messages being classified as German while actually also consisting

of a relevant fraction of e.g. English or other non-German languages. These mixed-language messages were filtered out later during the annotation process described in section However it could be argued that they should be included, since they represent realistic social media date and NER systems should be able to handle such kinds of texts as well.

As a preparatory step for the final sample selection, the dataset was reduced to only German-language texts (22.803 messages remaining). Out of these messages, only those with a minimum length of 100 characters (in the `cleaned_text` column) were selected (14.605 messages remaining).

An exploration of the text length of all German-language messages shows that the messages have a median text length of 207 characters, and 75% of the messages have a text length of up to 400 characters. The minimum text length is 5 characters, the longest text consists of 4458 characters.

	text_length
count	24001.0
mean	360.0
std	456.0
min	5.0
25%	125.0
50%	207.0
75%	403.0
max	4458.0

Figure 4.3: Descriptive statistiscs for text length of all German-language messages in dataset

Considering these insights, only messages consisting of at least 100 characters were selected (20.105 messages remaining). Out of these, twenty percent of the messages were randomly selected (4021 messages remaining).

Insights from prior research with Telegram data in the context of the project Digitaler Hass revealed that channels often contain a relatively high number of similar messages. To avoid redundancy in the sample, the similarity of the remaining messages was computed based on the `Levenshtein` distance:

```
import Levenshtein as lev
import pandas as pd
from datetime import datetime
from itertools import combinations


  def get_levenshtein_results(df, column='cleaned_text'):
    lev_df = pd.DataFrame()
    new_df = pd.DataFrame(combinations(df[column],2), columns
                                    =["text_1", "text_2"])
    new_df["lev_score"] = new_df.apply(lambda x: lev.ratio(x[
                                    0],x[1]), axis=1)
    lev_df = pd.concat([lev_df, new_df], axis=0).reset_index(
                                    drop=True)
    return lev_df


def get_message_ids(df, column='message_id'):
    lev_df_ids = pd.DataFrame()
    new_df_ids = pd.DataFrame(combinations(df[column],2),
                                    columns=["message_id_1", "
                                    message_id_2"])
    lev_df_ids = pd.concat([lev_df_ids, new_df_ids], axis=0).
                                    reset_index(drop=True)
    ts_after = datetime.now()
    return lev_df_ids
```

```
def combine_levenshtein_results_with_message_ids(lev_df,
                                lev_df_ids):
    levs = pd.DataFrame()
    levs = pd.concat([lev_df,lev_df_ids], axis=1)
    ts_after = datetime.now()
    return levs


def remove_similar_messages(lev_ids_df, sample_df, treshold):
    second_ids = lev_ids_df[lev_ids_df['lev_score'] >=
                                    treshold].message_id_2.
                                    tolist()
    ids_to_drop = list(set(second_ids))
    result_df = sample_df[~sample_df.message_id.isin(
                                    ids_to_drop)]
    return result_df
```

A treshold of 0.8 was defined. If the Levensthein score between for a pair of messages was greater than or equal to this treshold, the second message was removed from the sample.

The remaining sample after this step consisted of 3.095 messages.

In a second iteration, an additional sample of short texts was selected. This was supposed to reflect the fact that social media texts can be of very short length and thus should be included in the training data for NER systems. The sampling process for these short texts was similar to the first sampling iteration, except for the sampling size which was 50% instead of 20% in the first iteration. After random sampling and removing similar messages based on the Levenshtein score, a sample of 540 short messages remained and was added to the overall sample dataset.

Table 4.1: Annotation scheme: Entity types

Entity type	Description	Examples
PERSON	Named person or family, including artist names, names of fictional characters, etc.	
LOCATION	Countries, cities, states'; continents, districts, regions, streets and squares; shopping centers, parking lots, restaurants, 'Non-GPE locations, mountain ranges, bodies of water'; landmarks	
ORGANIZATION	Companies, agencies, institutions, newspapers, tv/radio channels, other media platforms; political parties; non-governmental organizations; political groups such as Querdenken or Antifa etc.	
DATE	Absolute or relative dates	23.09.2022
TIME	Specific times of day, time codes, periods of time; etc.	13:10
ACTION	Specific action or event mentioned in a text, e.g. demonstrations, pickets, motorcade; (protest) walks; etc. - must not be unique entity, plural allowed	
OTHER	Other entities, e.g. religions, nationalities, products, works of art, political spectres	

4.2 Annotation scheme

annotation schemes vary greatly in terms of: granularity, types of entities, detailedness of instructions and definitions

Loosely based on Benikova, Biemann, and Reznicek 2014 and spacy and on:

for the sake of simplicity: no nested entities, no derived entities

Only full noun phrases are considered as named entities. Pronouns and all other phrases can be ignored.

Names are denominators for a single unique object in the real world.

A named entity is a "real-world object" that's assigned a name – for example, a person, a country, a product or a book title. (spacy)

- NEs can consist of more than one token - Phrases only partly containing names (deutschlandweit), or adjectives referring to entities (euklidisch) can be ignored

combined NEs, e.g. Wien-Demo -¿ tagged separately if possible, otherwise tagged as the dominant entity type (here Demo), to avoid nested entities

labelstudio

IOB encoding: each word either begins, is inside, or is outside of a named entity

hashtags encoding events via location and date, e.g. #b20200809 or #le1311 are not to be annotated, should be extracted via regex and treated separately; unsure if they are NEs?

Table 4.2: Results from fine-tuning results different models on Telegram dataset

Model	**Validation Loss**	**F1 score**
dbmdz/bert-base-german-uncased	0.2285	0.7440

4.3 Annotated dataset

descriptive statistics max sequence length number of documents number of tokens per entity type

4.4 Baseline

Lit: Ritter et al. /Baldwin?

Create Baseline with XLM Roberta

4.5 Models

Experiments on smartdata, and Telegram separately: - DistilBERT - RoBERTa base - gbert base - dbmdz

selected: gbert and roberta base (best validation loss and F1)

4.6 Building a Pipeline

pipeline groups together three steps: preprocessing, passing the inputs through the model, and postprocessing

Like other neural networks, Transformer models can't process raw text directly, so the first step of our pipeline is to convert the text inputs into numbers that the model can make sense of. To do this we use a tokenizer, which will be responsible for:

- Splitting the input into words, subwords, or symbols (like punctuation) that are called tokens - Mapping each token to an integer - Adding additional inputs that may be useful to the model

4.6.1 Tokenization

All this preprocessing needs to be done in exactly the same way as when the model was pretrained

use Autotokenizer from_pretrained

Once we have the tokenizer, we can directly pass our sentences to it and we ll get back a dictionary that s ready to feed to our model

Subword tokenization, alignment

4.6.2 Vectorization?

convert the list of input IDs to tensors.

Transformer models only accept tensors as input. If this is your first time hearing about tensors, you can think of them as NumPy arrays instead. A NumPy array can be a scalar (0D), a vector (1D), a matrix (2D), or have more dimensions. It's effectively a tensor; other ML frameworks' tensors behave similarly, and are usually as simple to instantiate as NumPy arrays.

To specify the type of tensors we want to get back (PyTorch, TensorFlow, or plain NumPy), we use the return_tensors argument:

padding, truncation, return_tensors erklären können

Don t worry about padding and truncation just yet; we'll explain those later. The main things to remember here are that you can pass one sentence or a list of sentences, as well as specifying the type of tensors you want to get back (if no type is passed, you will get a list of lists as a result).

The output itself is a dictionary containing two keys, input_ids and attention_mask. input_ids contains two rows of integers (one for each sentence) that are the unique identifiers of the tokens in each sentence. We'll explain what the attention_mask is later in this chapter.

outputs

checkpoint / pretrained This architecture contains only the base Transformer module: given some inputs, it outputs what we'll call hidden states, also known as features. For each model input, we'll retrieve a high-dimensional vector representing the contextual understanding of that input by the Transformer model. While these hidden states can be useful on their own, they're usually inputs to another part of the model, known as the head.

The vector output by the Transformer module is usually large. It generally has three dimensions:

Batch size: The number of sequences processed at a time (2 in our example). Sequence length: The length of the numerical representation of the sequence (16 in our example). Hidden size: The vector dimension of each model input.

Note that the outputs of Transformers models behave like namedtuples or dictionaries. You can access the elements by attributes (like we did) or by key (outputs["last_hidden_state"]), or even by index if you know exactly where the thing you are looking for is (outputs[0]).

4.6.3 Model heads

The model heads take the high-dimensional vector of hidden states as input and project them onto a different dimension. They are usually composed of one or a few linear layers The output of the Transformer model is sent directly to the model head to be processed.

In this diagram, the model is represented by its embeddings layer and the subsequent layers. The embeddings layer converts each input ID in the tokenized input into a vector that represents the associated token. The subsequent layers manipulate those vectors using the attention mechanism to produce the final representation of the sentences.

There are many different architectures available in Transformers, with each one designed around tackling a specific task.

here: for token classification (NER)

4.6.4 Post-processing

The values we get as output from our model don't necessarily make sense by themselves

Those are not probabilities but logits, the raw, unnormalized scores outputted by the last layer of the model. To be converted to probabilities, they need to go through a SoftMax layer (all Transformers models output the logits, as the loss function for training will generally fuse the last activation function, such as SoftMax, with the actual loss function, such as cross entropy)

To get the labels corresponding to each position, we can inspect the id2label attribute of the model config

model.config.id2label

4.7 Evaluation

- training loss - validation loss - early stopping - F1 - precision, recall, accuracy erklären

Overfitting

Error Analysis

Confusion matrix

4.8 Hyperparameter tuning

4.9 Processing of results

4.10 Summarization

Chapter 5

Results

Chapter 6

Conclusion

6.1 Summarization

6.2 Discussion

more data

improved annotation scheme (zb ( ) / nicht taggen, # splitten), wie umgehen mit hashtags für Demos zB #b2908

entity disambiguation

relationship tagging

future work -¿ main adaptation on large telegram corpus

Bibliography

Aggarwal, Charu C. (2018). *Machine Learning for Text*. en. Cham: Springer International Publishing. ISBN: 978-3-319-73530-6 978-3-319-73531-3. DOI: 10.1007/978-3-319-73531-3. URL: http://link.springer.com/10.1007/978-3-319-73531-3 (visited on 05/14/2022).

Baldwin, Timothy et al. (July 2015). "Shared Tasks of the 2015 Workshop on Noisy User-generated Text: Twitter Lexical Normalization and Named Entity Recognition". In: *Proceedings of the Workshop on Noisy User-generated Text*. Beijing, China: Association for Computational Linguistics, pp. 126–135. DOI: 10.18653/v1/W15-4319. URL: https://aclanthology.org/W15-4319 (visited on 05/16/2022).

Benikova, Darina, Biemann, Chris, and Reznicek, Marc (May 2014). "NoSta-D Named Entity Annotation for German: Guidelines and Dataset". In: *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*. Reykjavik, Iceland: European Language Resources Association (ELRA), pp. 2524–2531. URL: http://www.lrec-conf.org/proceedings/lrec2014/pdf/276_Paper.pdf.

Benikova, Darina et al. (2014). "Germeval 2014 named entity recognition: Companion paper". In: *Proceedings of the KONVENS GermEval Shared Task on Named Entity Recognition, Hildesheim, Germany*, pp. 104–112.

Chinchor, Nancy and Sundheim, Beth (Aug. 2003). *Message Understanding Conference (MUC) 6*. Artwork Size: 10240 KB Pages: 10240 KB Type:

dataset. DOI: 10.35111/WBCC-Y063. URL: https://catalog.ldc.upenn.edu/LDC2003T13 (visited on 06/27/2022).

Collobert, Ronan et al. (2011). "Natural Language Processing (Almost) from Scratch". en. In: *NATURAL LANGUAGE PROCESSING*, p. 45.

Derczynski, Leon et al. (Mar. 2015). "Analysis of named entity recognition and linking for tweets". en. In: *Information Processing & Management* 51.2, pp. 32–49. ISSN: 03064573. DOI: 10.1016/j.ipm.2014.10.006. URL: https://linkinghub.elsevier.com/retrieve/pii/S0306457314001034 (visited on 06/27/2022).

Ethnologue (May 2022). *How many languages are there in the world?* en. URL: https://www.ethnologue.com/guides/how-many-languages (visited on 06/19/2022).

Grishman, Ralph and Sundheim, Beth (1996). "Message Understanding Conference-6: A Brief History". In: *COLING 1996 Volume 1: The 16th International Conference on Computational Linguistics*. URL: https://aclanthology.org/C96-1079 (visited on 06/27/2022).

Jurafsky, Daniel and Martin, James H. (2021). *Speech and Language Processing. An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*. URL: https://web.stanford.edu/~jurafsky/slp3/ed3book_jan122022.pdf.

Magueresse, Alexandre, Carles, Vincent, and Heetderks, Evan (June 2020). *Low-resource Languages: A Review of Past Work and Future Challenges*. Number: arXiv:2006.07264 arXiv:2006.07264 [cs]. URL: http://arxiv.org/abs/2006.07264 (visited on 06/19/2022).

Mikolov, Tomas et al. (2013). *Efficient Estimation of Word Representations in Vector Space*. DOI: 10.48550/ARXIV.1301.3781. URL: https://arxiv.org/abs/1301.3781.

Nadeau, David and Sekine, Satoshi (Aug. 2007). "A survey of named entity recognition and classification". en. In: *Lingvisticae Investigationes* 30.1, pp. 3–26. ISSN: 0378-4169, 1569-9927. DOI: 10.1075/li.30.1.03nad.

URL: http://www.jbe-platform.com/content/journals/10.1075/li.30.1.03nad (visited on 07/04/2022).

Ritter, Alan et al. (2011). "Named Entity Recognition in Tweets: An Experimental Study". In: *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. EMNLP '11. Edinburgh, United Kingdom: Association for Computational Linguistics, 1524–1534. ISBN: 9781937284114.

Schiersch, Martin et al. (2018). "A German Corpus for Fine-Grained Named Entity Recognition and Relation Extraction of Traffic and Industry Events". In: *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. Ed. by Nicoletta Calzolari (Conference chair) et al. Miyazaki, Japan: European Language Resources Association (ELRA). ISBN: 979-10-95546-00-9.

Schweter, Stefan and Akbik, Alan (2020). *FLERT: Document-Level Features for Named Entity Recognition*. arXiv: 2011.06993 [cs.CL].

Singh, Anil Kumar (2008). "Natural Language Processing for Less Privileged Languages: Where do we come from? Where are we going?" In: *Proceedings of the IJCNLP-08 Workshop on NLP for Less Privileged Languages*. URL: https://aclanthology.org/I08-3004 (visited on 06/19/2022).

Tjong Kim Sang, Erik F. (2002). "Introduction to the CoNLL-2002 Shared Task: Language-Independent Named Entity Recognition". In: *COLING-02: The 6th Conference on Natural Language Learning 2002 (CoNLL-2002)*. URL: https://aclanthology.org/W02-2024.

Tjong Kim Sang, Erik F. and De Meulder, Fien (2003). "Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition". In: *Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003*, pp. 142–147. URL: https://aclanthology.org/W03-0419 (visited on 06/19/2022).

Tunstall, Lewis et al. (2022). *Natural language processing with Transformers: building language applications with Hugging Face*. eng. First edition. Bei-

jing Boston Farnham Sebastopol Tokyo: O'Reilly. ISBN: 978-1-09-810324-8.

Vajjala, Sowmya et al. (2020). *Practical natural language processing: a comprehensive guide to building real-world NLP systems*. First edition. OCLC: on1125266646. Sebastopol, CA: O'Reilly Media. ISBN: 978-1-4920-5405-4.

Weichbrodt, Gregor and Stanjek, Grischa (2022). *Teledash – analysis and research software for Telegram*. URL: `https://github.com/democ-de/teledash`.

Yadav, Vikas and Bethard, Steven (Oct. 2019). *A Survey on Recent Advances in Named Entity Recognition from Deep Learning models*. Tech. rep. arXiv:1910.11470. arXiv:1910.11470 [cs] type: article. arXiv. URL: `http://arxiv.org/abs/1910.11470` (visited on 05/14/2022).

Zong, Chengqing, Xia, Rui, and Zhang, Jiajun (2021). "Information Extraction". en. In: *Text Data Mining*. Singapore: Springer Singapore, pp. 227–283. ISBN: 9789811600999 9789811601002. DOI: `10.1007/978-981-16-0100-2_10`. URL: `https://link.springer.com/10.1007/978-981-16-0100-2_10` (visited on 05/14/2022).

Appendix

List of Figures

List of Tables

List of Acronyms

Annotation guidelines

some sample text

Based on Benikova, Biemann, and Reznicek 2014 and spacy

and on:

for the sake of simplicity: no nested entities, no derived entities

Only full noun phrases are considered as named entities. Pronouns and all other phrases can be ignored.

Names are denominators for a single unique object in the real world.

A named entity is a "real-world object" that's assigned a name – for example, a person, a country, a product or a book title. (spacy)

- NEs can consist of more than one token - Phrases only partly containing names (deutschlandweit), or adjectives referring to entities (euklidisch) can be ignored

combined NEs, e.g. Wien-Demo -¿ tagged separately if possible, otherwise tagged as the dominant entity type (here Demo), to avoid nested entities

hier wird klar wie der shift von protest forms is: Achtung Demonstration wird Abgesagt ! Wir haben als Orga Team beschlossen , das wir keine Demos mehr anmelden werden ! Wir sagen klar Nein zu den Auflagen die uns vorgegeben werden . Deshalb , neuer Termin , wir fahren nach Plauen ! Lasst uns die Freie Sachsen und den Bürgermeisterkandidaten Thomas Kaden unterstützen ! 11062021 Raus auf die Straße

labelstudio

IOB encoding: each word either begins, is inside, or is outside of a named

Table 1: Annotation scheme: Entity types

Entity type	**Description**	**Examples**
PERSON	Named person or family, including artist names, names of fictional characters, etc.	
LOCATION	Countries, cities, states'; continents, districts, regions, streets and squares; shopping centers, parking lots, restaurants, 'Non-GPE locations, mountain ranges, bodies of water'; landmarks	
ORGANIZATION	Companies, agencies, institutions, newspapers, tv/radio channels, other media platforms; political parties; non-governmental organizations; political groups	
DATE	Absolute or relative dates	23.09.2022
TIME	Specific times of day, time codes	13:10
ACTION	Specific action or event mentioned in a text, e.g. demonstrations, pickets, motorcade; (protest) walks; etc. - must not be unique entity, plural allowed	
OTHER	Other entities, e.g. religions, nationalities, products, works of art, political spectres	

entity

hashtags encoding events via location and date, e.g. #b20200809 or #le1311 are not to be annotated, should be extracted via regex and treated separately; unsure if they are NEs?

Statutory Declaration

I herewith formally declare that I have written the submitted thesis independently. I did not use any outside support except for the quoted literature and other sources mentioned in the paper.
I clearly marked and separately listed all of the literature and all of the other sources which I employed when producing this academic work, either literally or in content.
I am aware that the violation of this regulation will lead to the failure of the thesis.

.....	.....
Name	Signature
.....	.....
Matriculation Number	Place and Date